

花老师班补充作业3

5. 选(B).

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| x^2 = 2x^2 < 1 \Rightarrow |x| < \frac{1}{\sqrt{2}} \Rightarrow R = \frac{1}{\sqrt{2}}$$

6. 选(C).

 $\sum a_n(x-1)^n$ 在 $x=-2$ 收敛 $\Rightarrow |x-1| < |-2-1| = 3$ 绝对收敛

$$\Leftrightarrow -2 < x < 4$$

 $\Rightarrow x=0$ 绝对收敛.

7. 选(D)

$$\lim_{n \rightarrow \infty} \sqrt[n]{|u_n|} = \lim_{n \rightarrow \infty} \frac{|x|}{\sqrt[n]{n} \cdot 3} = \frac{|x|}{3} < 1 \Rightarrow |x| < 3$$

 when $x=-3$, $\sum \frac{(-1)^n}{n}$ 条件收敛 $\Rightarrow [-3, 3)$
 $x=3$, $\sum \frac{1}{n}$ 发散
11. $\sum a_n x^n$ 在 $x=2$ 收敛 $\Rightarrow |x| < 2$ 绝对收敛 $\Rightarrow R=2$.

~~12. $\lim_{n \rightarrow \infty} \sqrt[n]{|u_n|} = \lim_{n \rightarrow \infty} \frac{2^n + (-3)^n}{2^{n+1} + (-3)^{n+1}}$~~

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{2^{n+1} + (-3)^{n+1}}{2^n + (-3)^n} \right| x^2 = \lim_{n \rightarrow \infty} \left| \frac{(-\frac{2}{3})^{n+1} + 1}{(-\frac{2}{3})^n + 1} \right| x^2 = \frac{x^2}{3} < 1 \Rightarrow |x| < \sqrt{3}$$

$$\Rightarrow R = \sqrt{3}$$

$$28. \lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{n(n+1)}{(n+1)(n+2)} \right| |x| = |x| < 1 \Rightarrow -1 < x < 1$$

 $x=-1$, $\sum \frac{(-1)^n}{n(n+1)}$ 绝对收敛 $\Rightarrow [-1, 1]$
 $x=1$, $\sum \frac{1}{n(n+1)}$ 绝对收敛

Record good times

证: $\sum_{n=1}^{\infty} \frac{x^n}{n(n+1)} = \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right) x^n = \sum_{n=1}^{\infty} \frac{x^n}{n} - \frac{1}{x} \sum_{n=1}^{\infty} \frac{x^{n+1}}{n+1}$ if $x \neq 0$.

$\left(\sum_{n=1}^{\infty} \frac{x^n}{n} \right)' = \sum_{n=1}^{\infty} x^{n-1} = \frac{1}{1-x} \Rightarrow \sum_{n=1}^{\infty} \frac{x^n}{n} = \int \frac{dx}{1-x} = -\ln(1-x) + C_1$
let $x=0 \Rightarrow C_1=0$

$\left(\sum_{n=1}^{\infty} \frac{x^{n+1}}{n+1} \right)' = \sum_{n=1}^{\infty} x^n = \frac{x}{1-x} \Rightarrow \sum_{n=1}^{\infty} \frac{x^{n+1}}{n+1} = \int \frac{x}{1-x} dx = \int \frac{1-(1-x)}{1-x} dx = -\ln(1-x) + C_2$
let $x=0 \Rightarrow C_2=0$
 $\Rightarrow -1 < x < 1$

if $x=1$, $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = S_n = 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \dots = 1 - \frac{1}{n+1} \rightarrow 1$
 $\Rightarrow \sum_{n=1}^{\infty} \frac{x^n}{n(n+1)} = \begin{cases} -\ln(1-x) + \frac{\ln(1-x)+x}{x}, & \text{if } x \neq 0 \\ 1, & x=1 \\ 0, & x=0 \end{cases}$ if $x \in (-1, 1)$

证: $\sum_{n=1}^{\infty} \frac{x^n}{n(n+1)} = \frac{1}{x} \sum_{n=1}^{\infty} \frac{x^{n+1}}{n(n+1)}$ if $x \neq 0$.

$\left(\sum_{n=1}^{\infty} \frac{x^{n+1}}{n(n+1)} \right)' = \sum_{n=1}^{\infty} \frac{x^n}{n}$, $\left(\sum_{n=1}^{\infty} \frac{x^n}{n} \right)' = \left(\sum_{n=1}^{\infty} x^{n-1} \right)' = \frac{1}{1-x}$
 $\Rightarrow \sum_{n=1}^{\infty} \frac{x^n}{n} = \int \frac{dx}{1-x} = -\ln(1-x) + C_1$
let $x=0 \Rightarrow C_1=0$

$\Rightarrow \sum_{n=1}^{\infty} \frac{x^{n+1}}{n(n+1)} = \int -\ln(1-x) dx = -x \ln(1-x) - \int x \cdot \frac{1}{1-x} dx = -x \ln(1-x) - \int \frac{1-(1-x)}{1-x} dx$
 $= -x \ln(1-x) + \ln(1-x) + x + C_2$ $-1 < x < 1$

let $x=0 \Rightarrow C_2=0$
 $\Rightarrow \sum_{n=1}^{\infty} \frac{x^n}{n(n+1)} = \begin{cases} -\ln(1-x) + \frac{\ln(1-x)}{x} + 1, & \text{if } x \neq 0 \\ 1, & x=1 \\ 0, & \text{if } x=0 \end{cases}$
if $x=0$, $\lim_{n \rightarrow \infty} \frac{|a_n|}{|a_{n+1}|} = 1$

35. $\sum a_n$ 绝对收敛 $\Rightarrow \lim_{n \rightarrow \infty} a_n = 0 \Rightarrow \exists \varepsilon_0, n > N, \forall n, |a_n| < \frac{1}{2} \Rightarrow -\frac{1}{2} < a_n < \frac{1}{2} \Rightarrow -\frac{1}{2} < a_{n+1} < \frac{1}{2}$